

Project Interim Report
On
Online Course Management System
(Using Core Java & OOP Concepts)

Object Oriented Programming with Java

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Abstract

The Online Course Management System is a console-based application developed using Core Java and Object-Oriented Programming (OOP) concepts. The main purpose of this project is to manage courses, student enrollments, and track student progress efficiently. In today's digital learning environment, managing multiple students and courses manually can be difficult and time-consuming. This system provides a simple solution to automate these tasks.

The project allows students to enroll in different courses, update their learning progress, and view course details. It uses classes, objects, and arrays to store and manage data. The system demonstrates how real-world e-learning platforms work in a simplified way. Overall, this project helps in understanding the practical implementation of OOP concepts like encapsulation, data management, and object interaction.

Introduction

The Online Course Management System is designed to simplify the process of managing courses and student information. It is a Java-based console application that focuses on implementing OOP concepts.

Objectives :

- Manage course records
- Manage student records
- Enroll students in courses
- Track student progress
- Reduce manual work
- Apply OOP concepts

This project simulates a basic online learning system where students can enroll and monitor their progress.

Scope

This project is suitable for small educational systems or learning environments. It can manage basic course and student data efficiently.

Future enhancements:

- Database integration
 - GUI interface using Java Swing
 - Online system access
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System Requirements

Hardware:

- Computer with 4GB RAM

Software:

- Java JDK
 - IDE (VS Code / Eclipse / IntelliJ)
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Methodology

The system works in a step-by-step process:

- First, courses and students are created
- Then, students enroll in courses
- After that, progress is updated
- Finally, all details are displayed

The project uses OOP concepts where:

- Classes represent real-world entities
 - Objects interact with each other
 - Arrays store multiple records
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Flowchart

System Flow

Start



Create Course



Create Student



Enroll Student



Update Progress



Display Details



End

Class Diagram

Course Class

- Variables:
 - courseId
 - courseName

- duration
- Methods:
 - getCourseDetails()

Student Class

- Variables:
 - studentName
 - studentId
 - enrolledCourses[]
 - progress[]
- Methods:
 - enroll()
 - updateProgress()
 - displayDetails()

Implementation

Classes Used

- Course Class → stores course details
- Student Class → manages enrollment and progress
- Main Class → controls execution

Logic Explanation

- Each student can enroll in multiple courses
- Courses are stored in arrays
- Progress is tracked using a parallel array
- Methods are used to update and display data

Sample Input/Output

Rahul enrolled in Java Programming
Priya enrolled in Java Programming

Student: Rahul

Java Programming - Progress: 80%

Student: Priya

Java Programming - Progress: 65%

Future Enhancements

- Use ArrayList instead of arrays
- Add search functionality
- Add menu-driven interface
- Add GUI using Java Swing

Gantt Chart

1. Planning & Structure

The project is divided into stages like planning, designing, coding, testing, and documentation. This helps in organizing work efficiently.

2. Time Management

Task	Start	End
Planning	Day 1	Day 1
Design	Day 2	Day 2
Coding	Day 3	Day 4
Testing	Day 5	Day 5
Documentation	Day 6	Day 6

This helps avoid delays and last-minute pressure.

Conclusion

The Online Course Management System successfully demonstrates the use of Java and OOP concepts in building a real-world application. It simplifies course and student management and provides an efficient way to track progress. The project also enhances programming skills and understanding of object interaction.

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